Action Archive

for BigFix Platform

Installation Guide

Version	1.1.0
Archive Server	Ubuntu Server 22.04 / 24.04 LTS
Audit Agent	BigFix Root Server (Windows)
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1. What's Included

The installation package includes a single file:

File	Size	Purpose	Install On
action-archive-latest.tar.gz	~8 MB	Archive Server application + installer	Ubuntu Server

> **Note:** The single installation file also contains the audit agent installer. The Audit Agent is optional but recommended. Without it, actions deleted from BigFix before they are collected will not be visible in the archive.

2. Architecture Overview

The diagram below illustrates how the components interact across your environment.

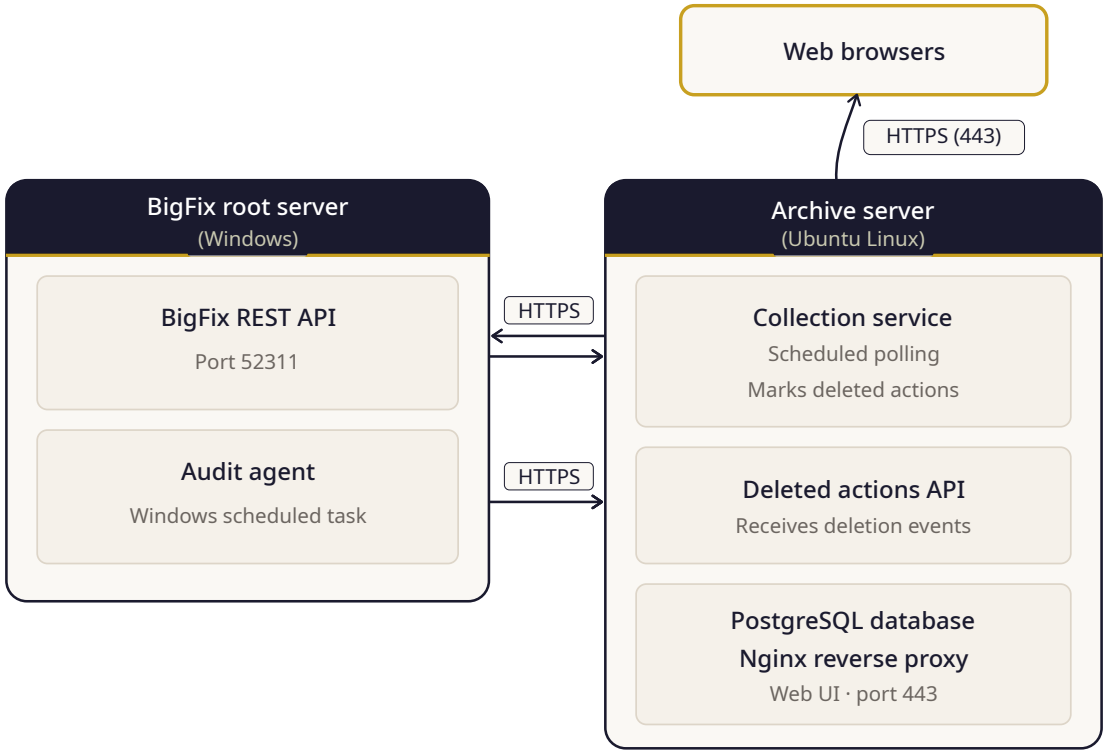


Figure 1 — Action Archive System Architecture

Data Flow

- **Collection Service:** The Archive Server polls the BigFix REST API on a schedule to collect expired and stopped actions. It can also be configured to mark collected actions as 'deleted' in the console for BigFix's cleanup process to remove from the database during a cleanup run.
- **Audit Agent:** Monitors server_audit.log on the BigFix Root Server and reports any action deletions to the Archive API via HTTPS.
- **Web UI:** Users access the archive through any web browser over HTTPS (port 443).

3. Prerequisites

Archive Server (Ubuntu)

Requirement	Details
Operating System	Ubuntu Server 22.04 LTS or 24.04 LTS (amd64)
Access	Root or sudo access
RAM	Minimum 4 GB
Disk	Minimum 20 GB free
Network	Internet access during installation (for apt packages)
Network	Must be able to reach the BigFix Root Server REST API
Ports	443 (HTTPS) must be accessible to users

> **Note:** The installer will automatically install and configure PostgreSQL, Nginx, Python, and all dependencies. You do not need to pre-install anything beyond the base Ubuntu Server.

BigFix Root Server (Windows) — Audit Agent

Requirement	Details
Operating System	Windows Server 2016+
Python	Python 3.8 or later (download from python.org)
PowerShell	5.1+ (included with Windows)
Network	Must be able to reach the Archive Server over HTTPS

BigFix Operator Account

The Archive Server requires a BigFix operator account to collect actions. This account must be a Master Operator (MO) to retrieve complete action data including targets and status.

> **Important:** Master Operator is the only role that provides complete access to all action data via the BigFix REST API. A non-MO operator will result in incomplete collections. It is recommended to create a dedicated service account for this purpose. The same requirement applies whether the installer is configured for Password or API Token authentication.

4. Part 1: Archive Server Installation (Ubuntu)

Step 1: Copy the Archive Package

Copy action-archive-latest.tar.gz to your Ubuntu server (e.g., /tmp folder):

Step 2: Extract the Package

```
ssh user@your-server
cd /tmp
tar -xzf action-archive-latest.tar.gz
cd action-archive
```

You should see the following files after extraction:

```
action-archive/
  app/                    # Application code
  static/                 # Web frontend files
  agent/                  # Bundled Audit Agent installer (Windows)
  install.sh              # Automated installer
  schema.sql              # Database schema
  requirements.txt
  README.md
  LICENSE
```

Step 3: Run the Installer

```
sudo bash install.sh
```

The installer will display a banner and begin the installation process.

Note: If an existing Action Archive installation is detected on this server, the installer presents an Upgrade / Reinstall / Cancel prompt. Upgrade (the default) preserves all configuration, credentials, and archived data and only refreshes the application files. Reinstall performs a fresh install and wipes existing data — use with caution. This prompt does not appear on a first-time install.

```
==============================================================================
  Action Archive for BigFix Platform — Installer
  Version 1.1.0
==============================================================================


[23:16:37] === Pre-flight Checks ===
  Detected: Ubuntu 24.04 (noble)
  Memory: 7941MB available
  Disk: 19958MB free on /opt
  WARNING: Action Archive for BigFix Platform is already installed (version: 1.1.0).

  Choose an option:
    1) Upgrade — replace application files, keep all config, credentials, users, and data (recommended)
    2) Reinstall — wipe configuration and start over (admin account and BigFix credentials will be re-prompted)
    3) Cancel

  Choice [1/2/3] (default 1):
```

Step 4: Enter Admin Account Details

The installer will prompt you for the initial administrator account:

- **Admin username:** The username you will use to log into the web UI (e.g., admin)
- **Admin password:** Must meet complexity requirements (minimum 8 characters, uppercase, lowercase, number, special character)
- **Admin email:** Email address for notifications and password resets

Step 5: Enter BigFix Connection Details

Next, the installer prompts for BigFix server connection information:

- **BigFix Server URL:** The hostname or IP of your BigFix Root Server (e.g., https://bigfix-server)
- **BigFix Server Port:** REST API port (default: 52311)
- **BigFix Authentication Method: Choose Password (default) or API Token. API Token is new in v1.1.0 and uses BigFix's native REST API token authentication (Authorization: Bearer header).**
- **If Password: enter the BigFix Operator Username (a Master Operator service account) and Password when prompted. If API Token: paste the token value when prompted (paste carefully — the token is encrypted at rest and cannot be retrieved later).**

Tip: To mint a BigFix API token, sign in to the BigFix Console as the operator, open Tools → Manage Tokens, and create a new token. The operator must retain the REST API permission. See the Administrator Guide for details.

```
--- BigFix Server Connection ---
BigFix server URL (e.g., https://bigfix.company.com): https://bfrootsvr
BigFix server port [52311]:
Authentication methods:
  1) Username + password (Master Operator)
  2) API token (BigFix 11.0.6+, operator needs 'Can use REST API' privilege)
Authentication method [1/2] [1]:
```

Step 6: SSL Certificate Configuration

The installer will ask about SSL certificates:

- **Option 1:** Generate a self-signed certificate (recommended for testing/internal use)
- **Option 2:** Provide paths to an existing certificate and key

For most installations, choose the self-signed option.

Step 7: Installation Completes

The installer will perform the following steps automatically:

- Install system packages (PostgreSQL, Nginx, Python, Node.js)
- Create the application user and directories
- Set up the Python virtual environment and install dependencies
- Initialize the database with the schema
- Configure Nginx as a reverse proxy with SSL
- Create and start the systemd service

When complete, you will see a success message with the URL to access the application.

5. Part 2: First Login and Configuration

Step 1: Open the Archive Server's Web UI

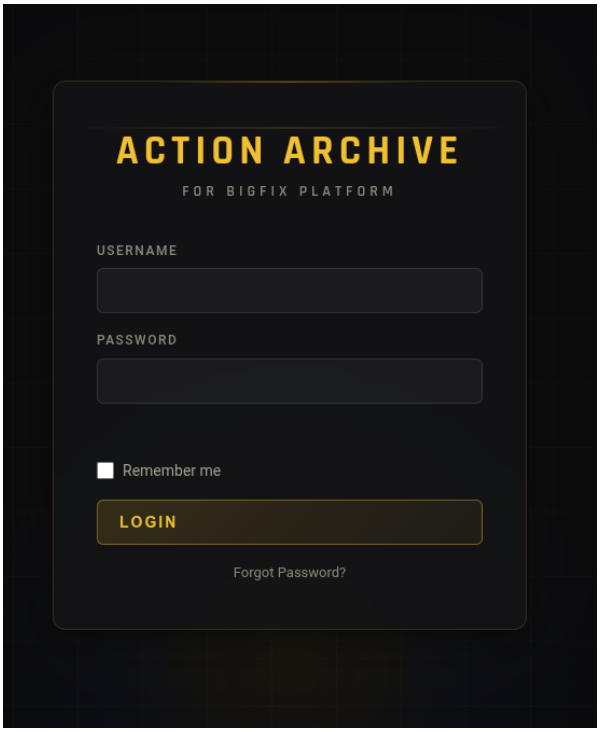
Open a web browser and navigate to the Archive Server URL:

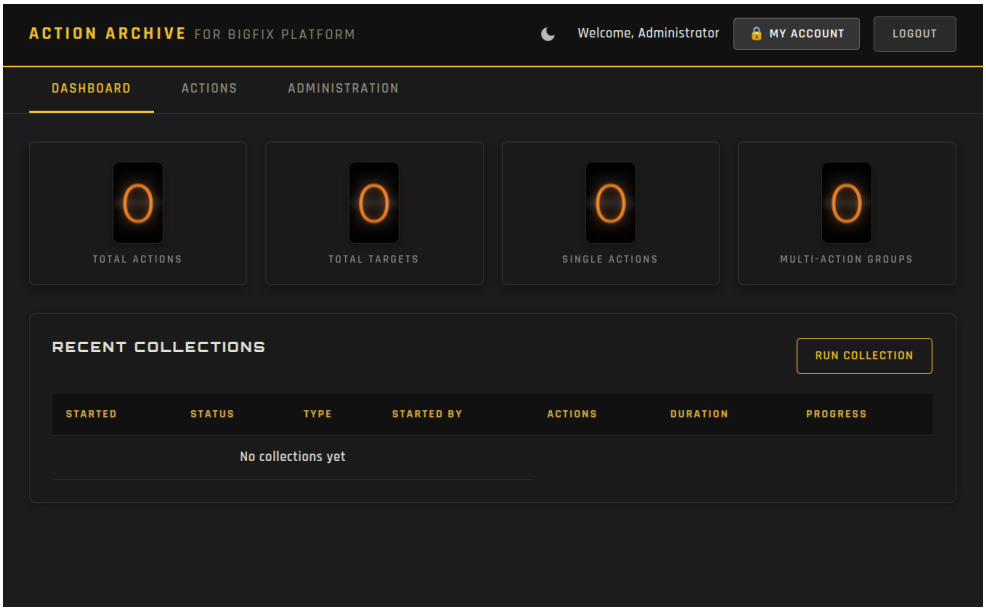
```
https://your-server-ip
```

> **Note:** If using a self-signed certificate, your browser will show a security warning.

Step 2: Log In

Enter the admin username and password you created during installation.

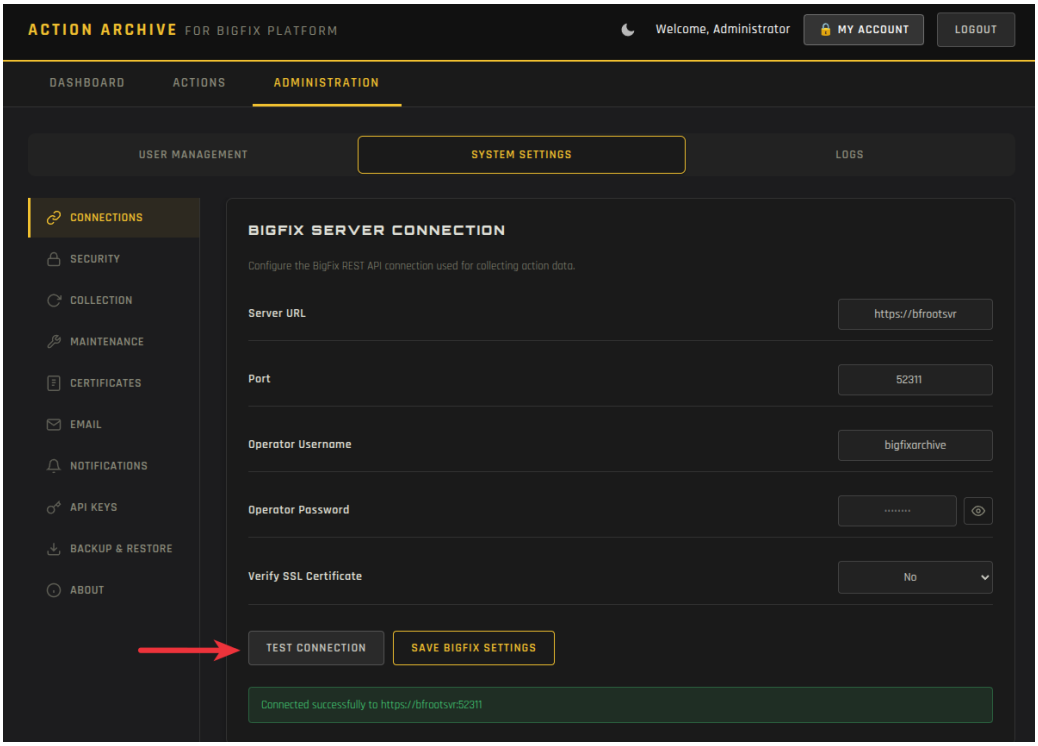




The dashboard should display the counters showing "0" for all statistics

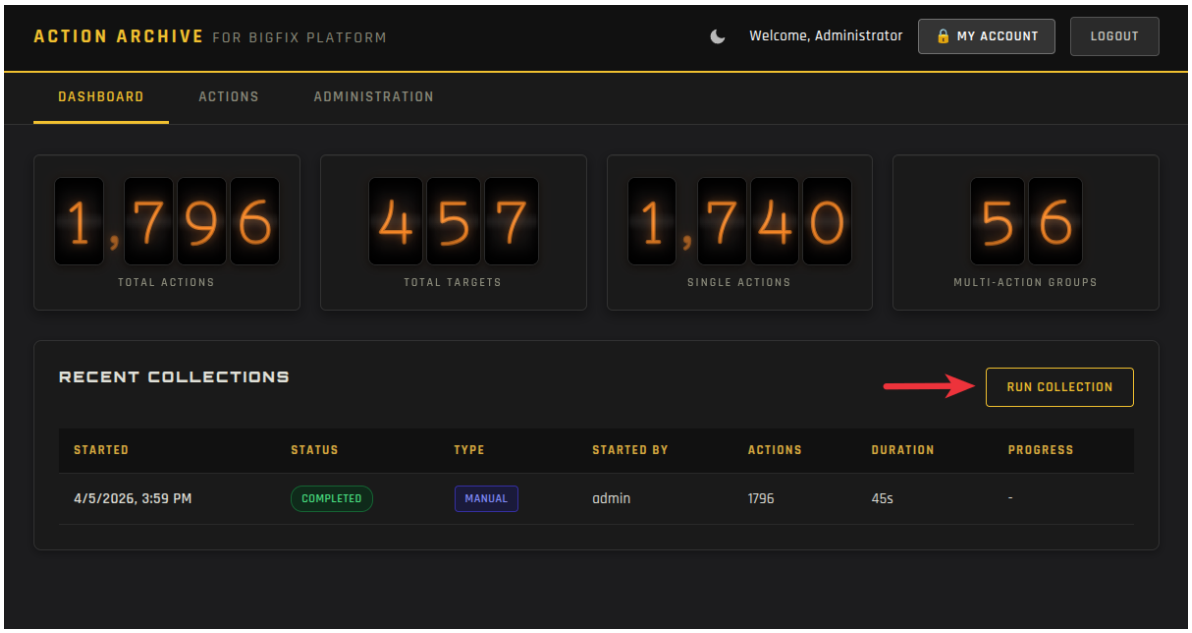
Step 3: Configure/Verify BigFix Connection

Navigate to Administration > System Settings > Connections to verify the BigFix server connection is correct. Click Test Connection to confirm the Archive can reach your BigFix REST API.



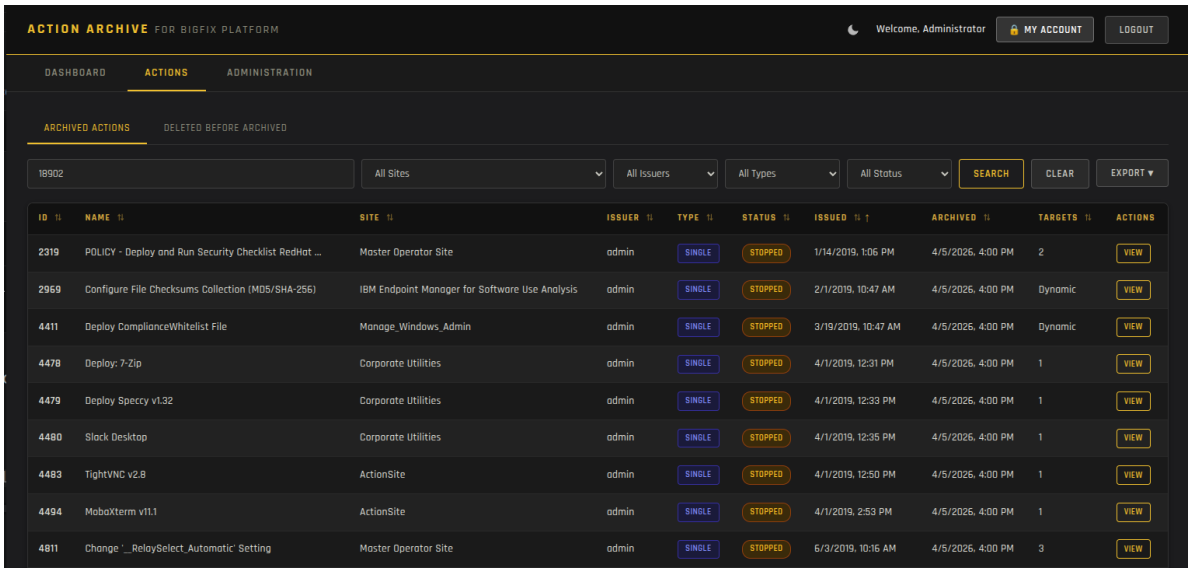
Step 4: Run Your First Collection

- Go to the Dashboard and click the **Run Collection** button to run an immediate collection

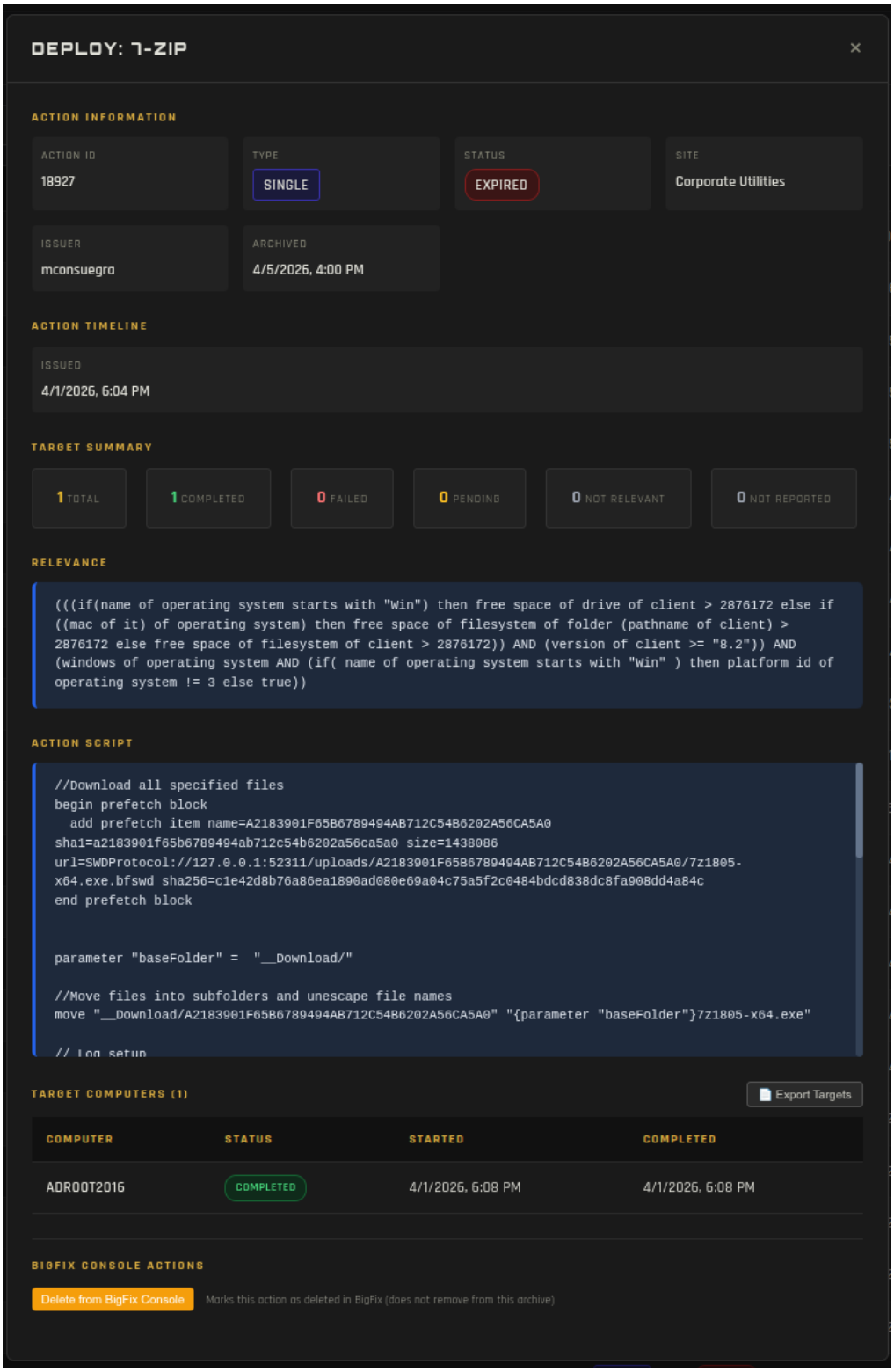


Step 5: Browse Archived Actions

After the first collection completes, navigate to the Actions tab to see your archived actions.



Click the 'View' button on an action to view the action's details.

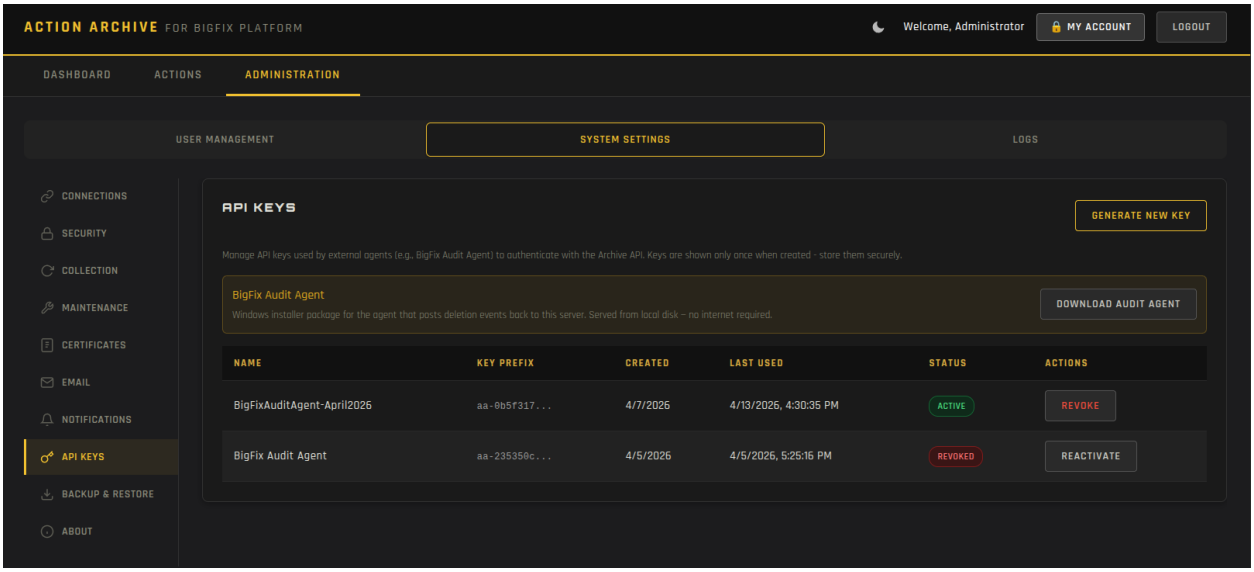


Step 6: Generate an API Key and Download the Audit Agent (Optional)

If you plan to install the Audit Agent (Part 3), both the API key and the agent installer package are obtained from the same panel of the web UI. No internet access is required — the agent installer is served from local disk on the Archive Server.

- Navigate to Administration > System Settings > API Keys
- Click Generate New Key
- Enter a name (e.g., "BigFix Audit Agent")
- Click Generate Key

Copy the generated key value immediately (it will not be shown again) and save it somewhere safe — you will paste it into the agent's installer in Part 3.
On the same API Keys panel, click Download Audit Agent to save action-archive-audit-agent-latest.zip to your local machine. You will transfer this file to the BigFix Root Server in Part 3.



> **Important:** Copy the displayed key immediately. It will only be shown once and cannot be retrieved later.

6. Part 3: Audit Agent Installation (Windows)

The Audit Agent is installed on the BigFix Root Server (Windows). It monitors the BigFix server_audit.log file for action deletions and reports them to the Archive Server.

Step 1: Install Python (if not already installed)

- Download Python 3.8 or later from python.org
- Run the installer

  IMPORTANT: Check the box **"Add python.exe to PATH"** on the first installer screen.

Step 2: Transfer the Audit Agent Package

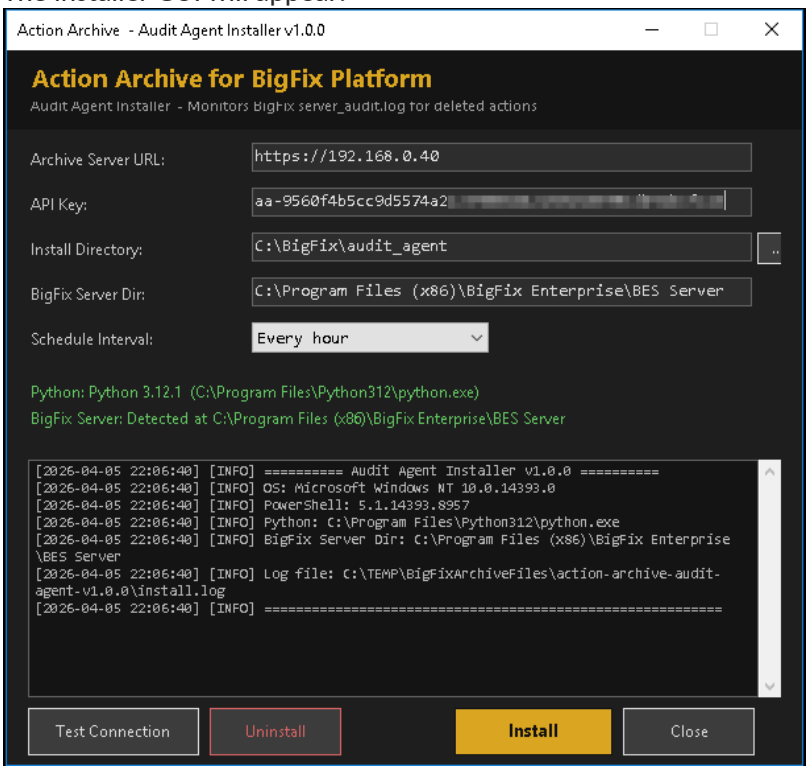
Transfer action-archive-audit-agent-latest.zip (downloaded in Part 2, Step 6) to the BigFix Root Server and extract it. You can place it anywhere (e.g., C:\Temp\ or the Desktop). If you did not download the agent in Part 2, sign in to the Archive web UI now and download it from Administration → System Settings → API Keys → Download Audit Agent.

Step 3: Run the Installer

- Navigate to the extracted folder
- **Double-click** Install.bat

> **Note:** If prompted by Windows SmartScreen or UAC, click "Run Anyway" / "Yes" to allow the installer to run. The installer requires Administrator privileges to create the scheduled task.

The installer GUI will appear:



Step 4: Enter Configuration

Fill in the following fields in the installer:

Field	Value	Notes
Archive Server URL	https://your-archive-server-ip	Must include https://
API Key	The key generated in Part 2, Step 6	Paste the full key starting with aa-
Install Directory	C:\BigFix\audit_agent	Default is fine for most environments
BigFix Server Dir	Auto-detected	Only change if BigFix is installed to a non-default location
Schedule Interval	Every hour	Configurable

Step 5: Test the Connection

Click Test Connection to verify the Archive Server URL and API Key are correct.

If the connection fails, verify:

- The Archive Server URL is correct and reachable from this server
- The API key was copied correctly (starts with aa-)
- The API key was generated on the Archive Server you are connecting to (keys are server-specific)

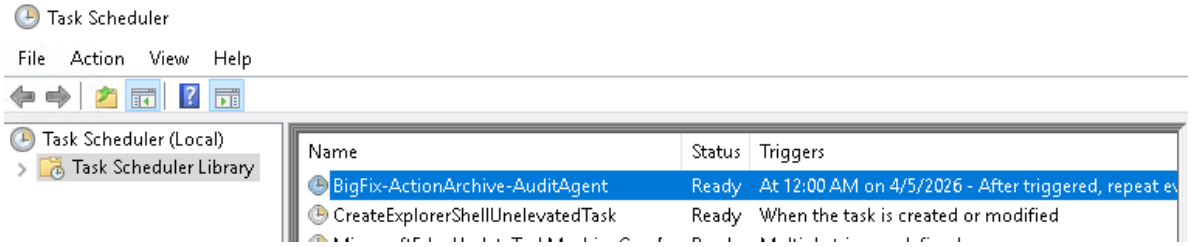
Step 6: Install

Click the Install button. The installer will perform the following steps:

- Create the install directory
- Copy agent scripts
- Generate config.ini with your settings
- Install the Python requests library
- Test the API connection
- Create a Windows Scheduled Task
- Run a historical scan of ALL server_audit.log files to catch past deletions
- Set the position marker for ongoing monitoring

Step 7: Confirm the Scheduled Task

Open Windows Task Scheduler and verify the task BigFix-ActionArchive-AuditAgent exists and is enabled.



7. Verification

After completing both installations, verify everything is working correctly.

Archive Server

1. Verify the service is running:

```
sudo systemctl status bigfix-archive
```

Should show active (running).

2. Check the health endpoint:

```
curl -sk https://localhost/api/health | python3 -m json.tool
```

Should return "status": "healthy" with all checks passing.

3. Web UI and data:

- Open https://your-server-ip in a browser and confirm you can log in
- Navigate to the Actions tab — it should show archived actions after a collection run

Audit Agent (Windows)

- **Scheduled task:** Open Task Scheduler and confirm BigFix-ActionArchive-AuditAgent is present
- **Agent log:** Check C:\BigFix\audit_agent\agent.log for successful runs
- **API Key activity:** In the Archive web UI, go to Settings > API Keys and verify the "Last Used" timestamp has updated
- **Deleted actions:** If any actions have been deleted from BigFix, they should appear in the Archive. You can test by deleting a test action from the BigFix Console, waiting for the next agent run, and checking the Archive.

8. Troubleshooting

Archive Server

Symptom	Likely Cause	Solution
Service fails to start	Configuration error	sudo journalctl -u bigfix-archive -n 50
"502 Bad Gateway" in browser	App not running or Nginx misconfigured	Verify service: sudo systemctl status bigfix-archive
Collection returns 0 actions	Operator is probably not a Master Operator	Verify MO status in BigFix Console
SSL certificate warning	Self-signed certificate in use	Expected for self-signed certs; proceed past the warning
Cannot reach BigFix server	Network or firewall issue	Verify: curl -sk https://bigfix-server:52311
Database connection error	PostgreSQL not running	sudo systemctl status postgresql

Audit Agent (Windows)

Symptom	Likely Cause	Solution
requests library not installed	pip failed during install	Run: pip install requests manually
Connection test fails	SSL or network issue	Verify Archive URL is reachable by opening it in a browser
API key rejected	Key mismatch or wrong server	Generate a new key on the correct Archive server
Agent runs but finds nothing	No deletions since last scan	Normal behavior — agent only reports new deletions
"Access denied" on scheduled task	Not running as Administrator	Right-click Install.bat > Run as Administrator
Historical scan found 0 deletions	No actions have been deleted	Normal if no one has deleted actions from BigFix

Log File Locations

Component	Log Location
Archive Server application	sudo journalctl -u bigfix-archive
Archive Server Nginx	/var/log/nginx/bigfix-archive-error.log
Audit Agent installer	install.log (in the extracted folder)
Audit Agent runtime	C:\BigFix\audit_agent\agent.log

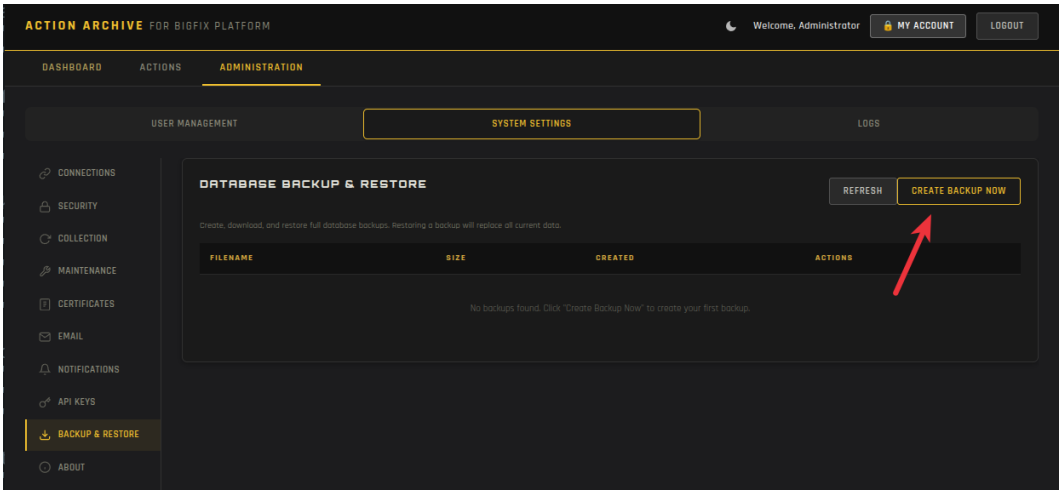
9. Managing the Application

Service Management (Ubuntu)

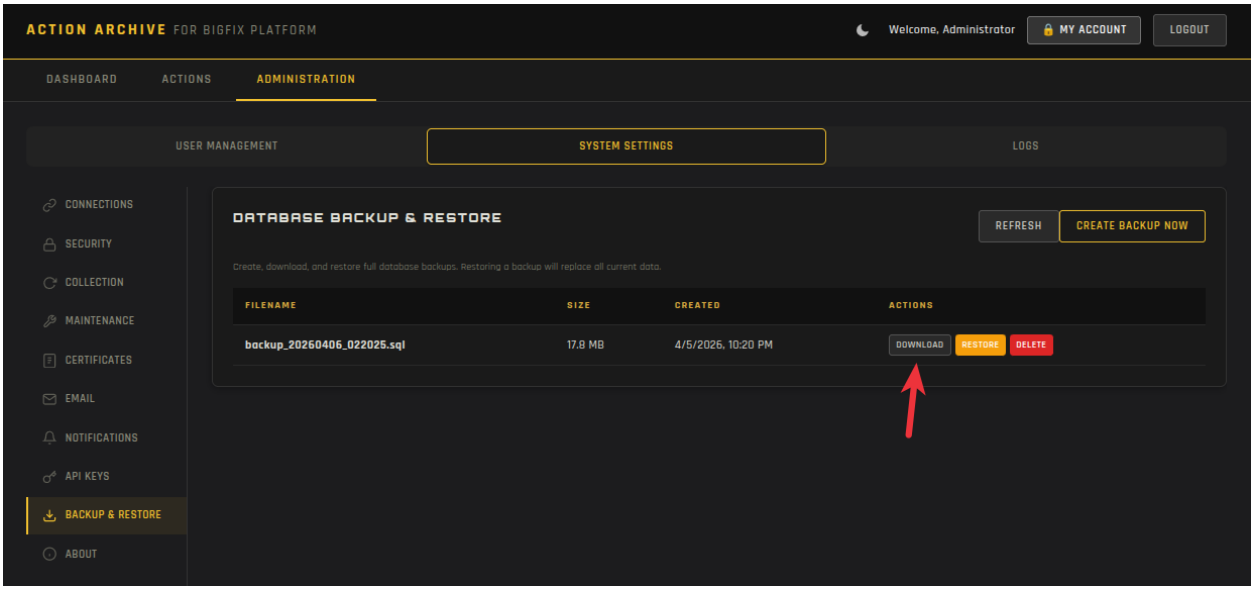
```
# Check status
sudo systemctl status bigfix-archive

# Restart the service
sudo systemctl restart bigfix-archive

# Stop the service
sudo systemctl stop bigfix-archive

# View recent logs
sudo journalctl -u bigfix-archive -n 100

# Follow logs in real-time
sudo journalctl -u bigfix-archive -f
```

Database Backup (from Web UI)

- Navigate to Administration > System Settings > Backup & Restore
- Click Create Backup



- Download the backup file for off-site storage



Audit Agent Management (Windows)

```
# View task status
Get-ScheduledTask -TaskName "BigFix-ActionArchive-AuditAgent"

# Run manually
Start-ScheduledTask -TaskName "BigFix-ActionArchive-AuditAgent"

# Disable temporarily
Disable-ScheduledTask -TaskName "BigFix-ActionArchive-AuditAgent"

# Re-enable
Enable-ScheduledTask -TaskName "BigFix-ActionArchive-AuditAgent"
```

Uninstalling the Audit Agent

Use the installer GUI: double-click Install.bat, then click the Uninstall button. This removes the scheduled task and all agent files.